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Microorganism identification method

Abstract

Provided is a method of identifying a serovar of *Salmonella* bacteria including a step of subjecting a sample containing microorganisms to mass spectrometry to obtain a mass spectrum, a step of reading a mass-to-charge ratio m/z of a peak derived from a marker protein from the mass spectrum, and an identification step of identifying a serovar of *Salmonella* bacteria in the sample, based on the mass-to-charge ratio m/z , wherein the serovars of *Salmonella* bacteria are classified using cluster analysis using as an index the mass-to-charge ratio m/z derived from at least 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, peptidylprolyl isomerase, gns, YibT, YaiA and YciF as the marker proteins.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Rule 53(b) Continuation application of U.S. application Ser. No. 16/089,836 filed Sep. 28, 2018, which is a National Stage of International Application No. PCT/JP2016/060865 filed Mar. 31, 2016, the respective disclosures of all of the above of which are incorporated herein by reference in their entirety.

INCORPORATION

(1) The content of the electronically submitted sequence listing, file name: Q289660_sequence listing as filed; size 1,232,443 bytes; and date of creation: Nov. 1, 2023, filed herewith, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present invention relates to a microorganism identification method using mass spectrometry.

BACKGROUND ART

(3) *Salmonella* belongs to the family of enterobacteriaceae of gram-negative facultative anaerobic bacilli, and three species of *Salmonella enterica*, *Salmonella bongori* and *Salmonella subterranea* belong to the genus *Salmonella*. Further, *Salmonella enterica* is classified into six subspecies (*Salmonella* (sometimes abbreviated as “S.”) *enterica* subsp. *enterica*, *S. enterica* subsp. *salamae*, *S. enterica* subsp. *arizonae*, *S. enterica* subsp. *diarizonae*, *S. enterica* subsp. *houtenae*, *S. enterica* subsp. *indica*).

(4) There are about 2,500 serovars in the genus *Salmonella*, which are decided by the Kauffmann-White classification based on the difference in combination of a cell wall lipopolysaccharide O antigen, and a flagellar protein H antigen. Pathogenic *Salmonella* such as *Salmonella* causing food poisoning belongs mostly to *S. enterica* subsp. *enterica*. This subspecies is also classified into about 1,500 types of serovars (Non Patent Literature 1). Currently, in order to decide the serovar, an agglutination test with antisera is used. It is an O type test by slide agglutination and an H type test by test tube agglutination, and the H type test increases mobility and performs phase induction for first phase and second phase decision, thus requires time and proficient skills for serovar decision.

(5) Some serovars have determined pathogenic hosts. For example, *typhi*, *choleraesuis*, Dublin and *gallinarum* cause systemic infection specifically in humans, pigs, cattle, and chickens. However, many other serovars infect multiple hosts like humans, domestic animals, pets and wild animals and become pathogens of nontyphoidal acute gastroenteritis (food poisoning). Infection routes of nontyphoidal *Salmonella* range widely such as environments such as rivers, wild animals, pets, and foods (including secondary pollution as well as primary pollution such as through rodents and insects). Serovar decision is important for infection prevention and epidemiological analysis and has been used for more than 80 years (Non Patent Literature 2).

(6) Highly detected serovars of nontyphoidal *Salmonella* infections in recent years are Enteritidis, Thompson, Infantis, Typhimurium, Saintpaul, Braenderup, Schwarzengrund, Litchfield, and Montevideo (IASR HP (Reference Document 1)). In the Act on Domestic Animal Infectious Diseases Control in Japan, when livestock is infected with Dublin, Enteritidis, Typhimurium or *choleraesuis*, notification to the Ministry of Agriculture, Forestry and Fisheries is mandatory.

(7) As methods for detecting *Salmonella* and deciding serovars, multiplex PCR (Non Patent Literatures 3 and 4), pulsed field gel electrophoresis (Non Patent Literature 5), multilocus sequence typing method (Non Patent Literature 6) and the like have been reported so far. However, with multiplex PCR, there are problems that only a few serovars are decided, or only a part of the O antigen and H antigen is decided, and the other methods require a complicated operation and take time.

(8) On the other hand, in recent years, the microorganism identification technique by matrix-assisted laser desorption/ionization time-of-flight mass-spectrometry (MALDI-TOF MS) has spread rapidly in clinical and food fields. This method is a method of identifying microorganisms based on a mass spectral pattern obtained using a very small amount of microorganism sample, which can obtain an analysis result in a short time and also easily perform continuous analysis of multiple specimens. Therefore, easy and rapid microorganism identification is possible. So far, attempts have been made to identify *Salmonella* using MALDI-TOF MS by multiple research groups (Non Patent Literatures 7, 8, 9, 10)

(9) Non Patent Literature 10 distinguishes subspecies of *Salmonella enterica* subsp. *enterica* and five major serovars by selecting a biomarker and preparing a decision tree. While the research by Dieckmann et al. scrutinizes protein peaks very minutely, there are strains in which biomarker peak is present or absent, and it takes time to confirm the peak.

CITATION LIST

Patent Literature

(10) Patent Literature 1: JP 2006-191922 A Patent Literature 2: JP 2013-085517 A

Non Patent Literature

(11) Non Patent Literature 1: ANTIGENIC FORMULAE OF THE SALMONELLA SEROVARS 2007 9th edition WHO Collaborating Center for Reference and Research on Salmonella Patrick A. D. Grimont, François-Xavier Weill Institut Pasteur, 28 rue du Dr. Roux, 75724 Paris Cedex 15, France Non Patent Literature 2: Winfield&Groisman, 2003, Fukuoka Institute of Health and Environmental Sciences Non Patent Literature 3: M Akiba Et. al., Microbiological Methods, 2011, 85, 9-15 Non Patent Literature 4: Y Hong et al., BMC microbiology 2008, 8: 178 Non Patent Literature 5: F Tenover, et al. Journal of clinical microbiology 33.9 (1995): 2233. Non Patent Literature 6: M Achtman, et al. PLoS Pathog 8.6 (2012): e1002776. Non Patent Literature 7: Seng, Piseth, et al. Future microbiology 5.11 (2010): 1733-1754. Non Patent Literature 8: M Kuhns et al. PLoS One 7.6 (2012): e40004. Non Patent Literature 9: R Dieckmann et al. AEM, 74.24 (2008): 7767-7778. Non Patent Literature 10: R Dieckmann, et al. (2011): AEM-02418. Non Patent Literature 11: T, Ojima-Kato, et al. PLOS one 2014: e113458.

SUMMARY OF INVENTION

Technical Problem

(12) On the other hand, Patent Literature 1 shows that a method (S10-GERMS method) of attributing the type of protein to be the origin of the peak by associating the mass-to-charge ratio of the peak obtained by mass spectrometry with a calculated mass estimated from the amino acid sequence obtained by translating the base sequence information of the ribosomal protein gene, utilizing the fact that about half of the peaks obtained by subjecting microbial cells to mass spectrometry is derived from ribosomal proteins, is useful (Patent Literature 1). According to this method, it is possible to perform highly reliable microorganism identification based on a theoretical basis using mass spectrometry and software attached thereto (Patent Literature 2).

(13) An object to be solved by the present invention is to provide a highly reliable biomarker based on genetic information that can rapidly and easily identify the serovar of *Salmonella enterica* subsp. *enterica*.

Solution to Problem

(14) As a result of extensive studies, the present inventors have found that two types of ribosomal proteins S8 and Peptidylpropyl isomerase are useful as marker proteins used for identifying which

- species of serovar of *Salmonella* genus bacteria is contained in a sample by mass spectrometry, and it is possible to identify the serovar of *Salmonella* genus bacteria reproducibly and quickly by using at least one of these ribosomal proteins, and have reached the present invention.
- (15) More specifically, a microorganism identification method according to the present invention, which has been made to solve the above problems, includes a) a step of subjecting a sample containing microorganisms to mass spectrometry to obtain a mass spectrum, b) a step of reading a mass-to-charge ratio m/z of a peak derived from a marker protein from the mass spectrum, and c) an identification step of identifying which bacteria of serovar of *Salmonella* genus bacteria the microorganisms contained in the sample contain, based on the mass-to-charge ratio m/z , in which at least one of two types of ribosomal proteins S8 and Peptidylpropyl isomerase is used as the marker protein.
- (16) In the above microorganism identification method, it is preferable that the serovars of *Salmonella* genus bacteria are classified using cluster analysis using as an index the mass-to-charge ratio m/z derived from at least 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, Peptidylpropyl isomerase, gns, YibT, YaiA and YciF as the marker protein.
- (17) In this case, it is preferable to further include a step of generating a dendrogram representing an identification result by the cluster analysis.
- (18) In addition, in the above microorganism identification method, when the serovar of *Salmonella* genus bacteria is Orion, at least Peptidylpropyl isomerase is preferably contained as the marker protein.
- (19) Moreover, when the serovar of *Salmonella* genus bacteria is Rissen, at least S8 is preferably contained as the marker protein.
- (20) Also, when the serovar of *Salmonella* genus bacteria is Saintpaul, at least L21, S7, YaiA and YciF are preferably contained as the marker protein.
- (21) Further, when the serovar of *Salmonella* genus bacteria is Braenderup, at least the group consisting of SOD, or gns and L25 is preferably contained as the marker protein.
- (22) Furthermore, when the serovar of *Salmonella* genus bacteria is Montevideo or Schwarzengrund, at least one of SOD and L21, and S7 are preferably contained as the marker protein.
- (23) Also, when the serovar of *Salmonella* genus bacteria is Enteritidis, at least SOD, L17 and S7 are preferably contained as the marker protein.
- (24) Further, when the serovar of *Salmonella* genus bacteria is *infantis*, at least SOD, L21, S7, YibT and YciF are preferably contained as the marker protein.

Advantageous Effects of Invention

- (25) According to the present invention, since a ribosomal protein showing a mutation peculiar to the serovar of *Salmonella* genus bacteria is used as the marker protein, the serovar of *Salmonella* genus bacteria can be reproducibly and quickly identified.
- (26) Also, by using a ribosomal protein showing a mutation peculiar to the serovar of *Salmonella* genus bacteria as the marker protein and performing a cluster analysis using the mass-to-charge ratio m/z of the peak derived from the marker protein on the mass spectrum as an index, the serovars of *Salmonella* genus bacteria contained in a plurality of samples can be collectively identified.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a configuration diagram showing a main part of a microorganism identification system used for a microorganism identification method according to the present invention.
- (2) FIG. 2 is a flowchart showing an example of a procedure of a microorganism identification

method according to the present invention.

(3) FIG. 3 shows a list of species name, subspecies name and serovar of *Salmonella* genus bacteria used in examples.

(4) FIG. 4 shows relationships between a combination of an agglutinated immune serum and a serovar.

(5) FIG. 5 shows a list of primers used in examples.

(6) FIG. 6 shows a mass of each amino acid.

(7) FIG. 7A shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 1).

(8) FIG. 7B shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 2).

(9) FIG. 7C shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 3).

(10) FIG. 7D shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 4).

(11) FIG. 7E shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 5).

(12) FIG. 7F shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 6).

(13) FIG. 7G shows a list of theoretical mass values of each ribosomal protein of *Salmonella* genus bacteria used in examples and measured values by MALDI-TOF MS (part 7).

(14) FIG. 8A is attribution results based on measured values of 12 types of ribosomal proteins (part 1).

(15) FIG. 8B is attribution results based on measured values of 12 types of ribosomal proteins (part 2).

(16) FIG. 8C is attribution results based on measured values of 12 types of ribosomal proteins (part 3).

(17) FIG. 8D is attribution results based on measured values of 12 types of ribosomal proteins (part 4).

(18) FIG. 9 is a chart obtained by MALDI-TOF MS measurement.

(19) FIG. 10A is identification results by SARAMIS (part 1).

(20) FIG. 10B is identification results by SARAMIS (part 2).

(21) FIG. 11 is a peak chart of ribosomal protein SOD.

(22) FIG. 12 is a peak chart of ribosomal protein L17.

(23) FIG. 13 is a peak chart of ribosomal protein L21.

(24) FIG. 14 is a peak chart of ribosomal protein S8.

(25) FIG. 15 is a peak chart of ribosomal protein L15.

(26) FIG. 16 is a peak chart of ribosomal protein S7.

(27) FIG. 17 is a peak chart of ribosomal protein gns.

(28) FIG. 18 is a peak chart of ribosomal protein YibT.

(29) FIG. 19 is a peak chart of ribosomal protein ppic.

(30) FIG. 20 is a peak chart of ribosomal protein L25.

(31) FIG. 21 is a peak chart of ribosomal protein YaiA.

(32) FIG. 22 is a peak chart of ribosomal protein YciF.

(33) FIG. 23 is a dendrogram generated using 12 types of ribosomal proteins.

(34) FIG. 24A is DNA sequences of ribosomal protein S8 (part 1).

(35) FIG. 24B is DNA sequences of ribosomal protein S8 (part 2).

(36) FIG. 24C is DNA sequences of ribosomal protein S8 (part 3).

(37) FIG. 24D is DNA sequences of ribosomal protein S8 (part 4).

(38) FIG. 25A is DNA sequences of ribosomal protein L15 (part 1).

(39) FIG. 25B is DNA sequences of ribosomal protein L15 (part 2).
(40) FIG. 25C is DNA sequences of ribosomal protein L15 (part 3).
(41) FIG. 25D is DNA sequences of ribosomal protein L15 (part 4).
(42) FIG. 25E is DNA sequences of ribosomal protein L15 (part 5).
(43) FIG. 26A is DNA sequences of ribosomal protein L17 (part 1).
(44) FIG. 26B is DNA sequences of ribosomal protein L17 (part 2).
(45) FIG. 26C is DNA sequences of ribosomal protein L17 (part 3).
(46) FIG. 26D is DNA sequences of ribosomal protein L17 (part 4).
(47) FIG. 26E is DNA sequences of ribosomal protein L17 (part 5).
(48) FIG. 27A is DNA sequences of ribosomal protein sodA (part 1).
(49) FIG. 27B is DNA sequences of ribosomal protein sodA (part 2).
(50) FIG. 27C is DNA sequences of ribosomal protein sodA (part 3).
(51) FIG. 27D is DNA sequences of ribosomal protein sodA (part 4).
(52) FIG. 27E is DNA sequences of ribosomal protein sodA (part 5).
(53) FIG. 27F is DNA sequences of ribosomal protein sodA (part 6).
(54) FIG. 27G is DNA sequences of ribosomal protein sodA (part 7).
(55) FIG. 28A is DNA sequences of ribosomal protein L21 (part 1).
(56) FIG. 28B is DNA sequences of ribosomal protein L21 (part 2).
(57) FIG. 28C is DNA sequences of ribosomal protein L21 (part 3).
(58) FIG. 28D is DNA sequences of ribosomal protein L21 (part 4).
(59) FIG. 29A is DNA sequences of ribosomal protein L25 (part 1).
(60) FIG. 29B is DNA sequences of ribosomal protein L25 (part 2).
(61) FIG. 29C is DNA sequences of ribosomal protein L25 (part 3).
(62) FIG. 30A is DNA sequences of ribosomal protein S7 (part 1).
(63) FIG. 30B is DNA sequences of ribosomal protein S7 (part 2).
(64) FIG. 30C is DNA sequences of ribosomal protein S7 (part 3).
(65) FIG. 30D is DNA sequences of ribosomal protein S7 (part 4).
(66) FIG. 30E is DNA sequences of ribosomal protein S7 (part 5).
(67) FIG. 31A is DNA sequences of ribosomal protein gns (part 1).
(68) FIG. 31B is DNA sequences of ribosomal protein gns (part 2).
(69) FIG. 32A is DNA sequences of ribosomal protein yibT (part 1).
(70) FIG. 32B is DNA sequences of ribosomal protein yibT (part 2).
(71) FIG. 33A is DNA sequences of ribosomal protein ppiC (part 1).
(72) FIG. 33B is DNA sequences of ribosomal protein ppiC (part 2).
(73) FIG. 34 is DNA sequences of ribosomal protein yaiA.
(74) FIG. 35A is DNA sequences of ribosomal protein yciF (part 1).
(75) FIG. 35B is DNA sequences of ribosomal protein yciF (part 2).
(76) FIG. 36A is amino acid sequences of ribosomal protein SOD (part 1).
(77) FIG. 36B is amino acid sequences of ribosomal protein SOD (part 2).
(78) FIG. 36C is amino acid sequences of ribosomal protein SOD (part 3).
(79) FIG. 37A is amino acid sequences of ribosomal protein L17 (part 1).
(80) FIG. 37B is amino acid sequences of ribosomal protein L17 (part 2).
(81) FIG. 38A is amino acid sequences of ribosomal protein L21 (part 1).
(82) FIG. 38B is amino acid sequences of ribosomal protein L21 (part 2).
(83) FIG. 39A is amino acid sequences of ribosomal protein S8 (part 1).
(84) FIG. 39B is amino acid sequences of ribosomal protein S8 (part 2).
(85) FIG. 40A is amino acid sequences of ribosomal protein L15 (part 1).
(86) FIG. 40B is amino acid sequences of ribosomal protein L15 (part 2).
(87) FIG. 41A is amino acid sequences of ribosomal protein S7 (part 1).
(88) FIG. 41B is amino acid sequences of ribosomal protein S7 (part 2).

- (89) FIG. **42** is amino acid sequences of ribosomal protein gns.
(90) FIG. **43** is amino acid sequences of the ribosomal protein YibT.
(91) FIG. **44** is amino acid sequences of the ribosomal protein ppic.
(92) FIG. **45** is amino acid sequences of ribosomal protein L25.
(93) FIG. **46** is amino acid sequences of ribosomal protein YaiA.
(94) FIG. **47** is amino acid sequences of ribosomal protein YciF.

DESCRIPTION OF EMBODIMENTS

(95) Hereinafter, a specific embodiment of the microorganism identification method according to the present invention will be described.

(96) FIG. **1** is an overview of a microorganism identification system used for a microorganism identification method according to the present invention. This microorganism identification system is roughly composed of a mass spectrometry unit **10** and a microorganism discrimination unit **20**. The mass spectrometry unit **10** includes an ionization section **11** for ionizing molecules and atoms in a sample by a matrix-assisted laser desorption ionization (MALDI) method, a time-of-flight mass separator (TOF) **12** for separating various kinds of ions emitted from the ionization section **11** according to the mass-to-charge ratio.

(97) The TOF **12** includes an extraction electrode **13** for extracting ions from the ionization section **11** and leading the ions to an ion flight space in the TOF **12**, and a detector **14** for detecting ions mass-separated in the ion flight space.

(98) The substance of the microorganism discrimination unit **20** is a computer such as a workstation or a personal computer, in which a Central Processing Unit (CPU) **21** that is a central processing unit, a memory **22**, a display section **23** consisting of a Liquid Crystal Display (LCD) and the like, an input section **24** consisting of a keyboard, a mouse and the like, and a storage section **30** consisting of a mass storage device such as a hard disk and a SSD (Solid State Drive) are connected to each other. In the storage section **30**, an Operating System (OS) **31**, a spectrum generation program **32**, a genus/species decision program **33**, and a subclass decision program **35** (program according to the present invention) are stored, and also a first database **34** and a second database **36** are housed. The microorganism discrimination unit **20** further includes an interface (I/F) **25** for direct connection with an external device and for controlling connection with an external device or the like via a network such as a LAN (Local Area Network), and is connected to the mass spectrometry unit **10** via a network cable NW (or wireless LAN) from the interface **25**.

(99) In FIG. **1**, the spectrum acquisition part **37**, the m/z reading part **38**, the subclass determination part **39**, the cluster analysis part **40**, and the dendrogram (system diagram) generation part **41** are shown as related with the subclass decision program **35**. Basically, these are all functional means realized by software by the CPU **21** executing the subclass decision program **35**. The subclass decision program **35** is not necessarily a single program but may be a function incorporated in a part of a program for controlling the genus/species decision program **33** or the mass spectrometry unit **10**, for example, and its form is not particularly limited. As the genus/species decision program **33**, for example, a program for performing microorganism identification by a conventional fingerprint method or the like can be used.

(100) Also, in FIG. **1**, a configuration in which the spectrum generation program **32**, the genus/species decision program **33**, and the subclass decision program **35**, the first database **34**, and the second database **36** are mounted on the terminal operated by the user is shown. However, a configuration in which at least part or all of them is provided in another device connected to the terminal via the computer network, and processing according to a program provided in the another device and/or access to the database is executed according to an instruction from the terminal may be used.

(101) A large number of mass lists related to known microorganisms are registered in the first database **34** of the storage section **30**. This mass list lists the mass-to-charge ratios of ions detected upon mass spectrometry of certain microbial cells. In addition to the information of the mass-to-

charge ratio, at least, information (classification information) of the classification group (family, genus, species, etc.) to which the microbial cells belong is contained. Such mass list is desirably created on the basis of data (measured data) obtained by actually subjecting various microbial cells to mass spectrometry in advance by the same ionization method and mass separation method as those by the mass spectrometry unit **10**.

(102) When creating a mass list from the measured data, first, a peak appearing in a predetermined mass-to-charge ratio range is extracted from the mass spectrum acquired as the measured data. At this time, by setting the mass-to-charge ratio range to about 2,000 to 35,000, it is possible to mainly extract a protein-derived peak. Also, by extracting only peaks whose height (relative intensity) is equal to or greater than a predetermined threshold, undesirable peaks (noise) can be excluded. Since the ribosomal protein group is expressed in a large amount in the cell, most of the mass-to-charge ratio described in the mass list can be derived from the ribosomal protein by appropriately setting the threshold. Then, the mass-to-charge ratios (m/z) of the peaks extracted as above are listed for each cell and registered in the first database **34** after adding the classification information and the like. In order to suppress variations in gene expression due to culture conditions, it is desirable to standardize culture conditions in advance for each microbial cell used for collecting the measured data.

(103) In the second database **36** of the storage section **30**, information on marker proteins for identifying known microorganisms by a classification (subspecies, pathotype, serovar, strain, etc.) lower than the species is registered. Information on the marker protein includes at least information on the mass-to-charge ratio (m/z) of the marker protein in the known microorganisms. In the second database **36** in the present embodiment, the values of mass-to-charge ratio m/z derived from at least 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, Peptidylpropyl isomerase, gns, YibT, YaiA and YciF are stored, as information on a marker protein for determining which serovar of *Salmonella* genus bacteria a test microorganism is. The values of mass-to-charge ratio of these ribosomal proteins will be described later.

(104) It is desirable that the values of mass-to-charge ratio of the marker protein stored in the second database **36** are selected by comparing the calculated mass obtained by translating the base sequence of each marker protein into an amino acid sequence with the mass-to-charge ratio detected by actual measurement. The base sequence of the marker protein can be decided by sequence, or also can use a public database, for example, one acquired from a database of NCBI (National Center for Biotechnology Information) or the like. When obtaining the calculated mass from the above amino acid sequence, it is desirable to consider cleavage of the N-terminal methionine residue as a post-translational modification. Specifically, when the penultimate amino acid residue is Gly, Ala, Ser, Pro, Val, Thr or Cys, the theoretical value is calculated assuming that the N-terminal methionine is cleaved. In addition, since molecules added with protons are actually observed by MALDI-TOF MS, it is desirable to obtain the calculated mass also considering the protons (that is, the theoretical value of mass-to-charge ratio of ions obtained when each protein is analyzed by MALDI-TOF MS).

(105) The procedure for identifying the serovar of *Salmonella* genus bacteria using the microorganism identification system according to this embodiment will be described with reference to a flowchart.

(106) First, the user prepares a sample containing constituents of test microorganism, sets the sample in the mass spectrometry unit **10**, and performs mass spectrometry. At this time, as the sample, in addition to a cell extract, or a cellular constituent such as a ribosomal protein purified from a cell extract, a bacterial cell or a cell suspension can be also used as it is.

(107) The spectrum generation program **32** acquires a detection signal acquired from the detector **14** of the mass spectrometry unit **10** via the interface **25**, and generates a mass spectrum of the test microorganism based on the detection signal (Step **S101**).

(108) Next, the species decision program **33** collates the mass spectrum of the test microorganism

with the mass lists of the known microorganisms recorded in the first database **34**, and extracts a mass list of the test microorganism having a mass-to-charge ratio pattern similar to the mass spectrum of the test microorganism, for example, a mass list containing many peaks that coincide with each peak in the mass spectrum of the test microorganism in a predetermined error range (Step **S102**). The species decision program **33** subsequently refers to the classification information stored in the first database **34** in association with the mass list extracted in Step **S102** to specify a species to which the known microorganism corresponding to the mass list belongs (Step **S103**). Then, when this species is not *Salmonella* genus bacteria (No in Step **S104**), the species is outputted to the display section **23** as a species of the test microorganism (Step **S116**), and the identification processing is terminated. On the other hand, when the species is *Salmonella* genus bacteria (Yes in Step **S104**), then the process proceeds to the identification processing by the subclass decision program **35**. When it is determined in advance that the sample contains *Salmonella* genus bacteria by other methods, the process may proceed to the subclass decision program **35** without utilizing the species decision program using the mass spectrum.

(109) In the subclass decision program **35**, first, the subclass determination part **39** reads out each of the values of mass-to-charge ratio of 12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, Peptidylpropyl isomerase, gns, YibT, YaiA and YciF from the second database **36** (Step **S105**). Subsequently, the spectrum acquisition part **37** acquires the mass spectrum of the test microorganism generated in Step **S101**. Then, the m/z reading part **38** selects peaks appearing in the mass-to-charge ratio range stored in the second database **36** in association with each marker protein on the mass spectrum as peaks corresponding to each marker protein, and reads the mass-to-charge ratio (Step **S106**). And, cluster analysis using the read mass-to-charge ratio as an index is performed. Specifically, the subclass determination part **39** compares the mass-to-charge ratio with the values of mass-to-charge ratio of each marker protein read out from the second database **36** and decides attribution of the protein with respect to the read mass-to-charge ratio (Step **S107**). Then, cluster analysis is performed based on the decided attribution to determine the serovar of the test microorganism (Step **S108**), and the result is output to the display section **23** as the identification result of the test microorganism (Step **S109**).

(110) Although the embodiments for carrying out the present invention have been described above with reference to the drawings, the present invention is not limited to the above-described embodiments, and appropriate modifications are permitted within the scope of the gist of the present invention.

EXAMPLES

(111) (1) Strains Used

(112) As described in FIG. 3, a total of 64 strains of *Salmonella* available from the National Institute of Technology and Evaluation Nite Biological Resource Center (NBRC), Microbe Division/Japan Collection of Microorganisms (JCM) RIKEN BioResource Research Center (Tsukuba), National Bioresource Project GTC Collection (Gifu) and the American Type Culture Collection (Manassas, VA, USA) that are strain culture collection, isolates from Japan Food Research Laboratories and isolates from Hyogo Prefectural Institute of Public Health science were used for analysis. The serovar of *Salmonella enterica* subsp. *enterica* was decided by multiplex PCR method reported by *Salmonella* immune serum “Seiken” (DENKA SEIKEN Co., Ltd.) and Non Patent Literatures 3 and 4. The strains were classified into 22 serovars by this method. FIG. 4 shows relationships between O-antigen immune serum and a serovar.

(113) (2) Analysis of DNA

(114) Among the primers used in *Escherichia coli* database creation (Non Patent Literature 11), those which cannot be shared with *Salmonella* genus bacteria were designed based on consensus sequences. The designed primers are shown in FIG. 5. Using these primers, DNA sequences of S10-spc-alpha operon and protein genes that could be biomarkers were analyzed. Specifically, genomic extraction was performed from each strain by a conventional method, and PCR was

carried out using KOD plus as a template to amplify a target gene region. The obtained PCR product was purified and used as a template for sequence analysis. Sequence analysis was performed using Big Dye ver. 3.1 Cycle Sequencing Kit (Applied Biosystems, Foster City, California, USA). The DNA sequence of the gene was converted to the amino acid sequence of each gene, and the mass-to-charge ratio was calculated based on the amino acid mass in FIG. 6 to obtain a theoretical mass value.

(115) (3) Analysis by MALDI-TOF MS

(116) Bacterial cells grown in Luria Agar medium (Sigma-Aldrich Japan, Tokyo, Japan) were recovered and approximately 2 colonies of bacterial cells were added in 10 μ L of a sinapinic matrix agent (25 mg/mL sinapinic acid (Wako Pure Chemical Industries, Ltd., Osaka, Japan) in 50 v/v % acetonitrile and 0.6 v/v % trifluoroacetic acid solution) and stirred well, and 1.2 μ L out of the solution was loaded on a sample plate and air-dried. For MALDI-TOF MS measurement, the sample was measured in positive linear mode, at spectral range of 2000 m/z to 35000 m/z using AXIMA microorganism identification system (Shimadzu Corporation, Kyoto City, Japan). The above-described calculated mass was matched with the measured mass-to-charge ratio with a tolerance of 500 ppm, and proper modification was made. The calibration of the mass spectrometer was performed according to the instruction manual, using *Escherichia coli* DH5a strain.

(117) (4) Construction of *Salmonella enterica* Subsp. *enterica* Database

(118) By comparing the theoretical mass values of the ribosomal proteins obtained in the above (2) with the peak chart by MALDI-TOF MS obtained in (3), it was confirmed that there was no difference between the theoretical values obtained from gene sequences and the measured values, regarding the protein which could be detected by actual measurement. The theoretical and measured values of the ribosomal proteins in the S10-spc- α operon and proteins that can be other biomarkers showing different masses depending on the strain are summarized as a database as shown in FIGS. 7A to 7G.

(119) The numbers shown in FIGS. 7A to 7G are the theoretical mass of the mass-to-charge ratio (m/z) obtained from genes. In addition, symbols “○”, “Δ”, and “x” represent mass peak detection results in actual measurement. Specifically, the symbol “○” indicates that it was detected as a peak within the 500 ppm range of the theoretical value at the default peak processing setting (threshold offset; 0.015 mV, threshold response; 1.200) of AXIMA microorganism identification system, and the symbol “x” indicates that there was a case where a peak could not be detected. In addition, the symbol “Δ” means that the theoretical mass difference in each strain or the difference from other protein peaks was 500 ppm or less, respectively, and the mass difference could not be identified even when a peak was detected.

(120) As can be seen from FIGS. 7A to 7G, it was showed that the theoretical mass values of the ribosomal proteins L23, L16, L24, S8, L6, S5, L15 and L17 encoded in the s10-spc-alpha operon and L21, L25, S7, SODa, gns, YibT, Peptidylpropyl isomerase, YaiA and YciF outside the operon (total 17 types) differ depending on the strain of *Salmonella enterica* subsp. *enterica*, thus are possibly useful protein markers that can be used for serovar identification of *Salmonella enterica* subsp. *enterica*.

(121) However, while it can be seen that L23, L16, L24, L6 and S5 have strains whose theoretical mass differences are separated by 500 ppm or more and can be a powerful biomarker for identification of these strains, there was a strain that could not be detected in actual measurement.

(122) On the other hand, a total of seven types of proteins, S8, L15, L17, L21, L25, S7 and Peptidylpropyl isomerase, were stably detected irrespective of the strains, and the mass difference by the strains was also 500 ppm or more. Therefore, these proteins were found useful as biomarkers for serovar identification of *Salmonella enterica* subsp. *enterica* in MALDI-TOF MS.

(123) SODa is an important biomarker for serovar identification of *Salmonella enterica* subsp. *enterica*, but the genotypes were varied and seven different mass-to-charge ratios were confirmed. All of these mass-to-charge ratios are as large as m/z around 23000, and in this region, the analysis

accuracy of currently provided MALDI-TOF MS is low unless the difference between the other mass-to-charge ratios is 800 ppm or more, thus SODa cannot identify the serovars. Therefore, four types that can identify the serovar at this time were used as biomarkers. Regarding gns, YibT, YaiA and YciF, contamination peaks exist in one of the theoretical mass values, but since serovars *infantis*, Thompson and Typhimurium are proteins that are mutated specifically, only the theoretical mass value without contamination peak was used as a biomarker. Therefore, 12 types of proteins were used as biomarkers for *Salmonella enterica* subsp. *enterica* serovar identification.

(124) (5) Attribution of Measured Values of MALDI-TOFMS by Software

(125) Based on the above, using a total of 12 types of proteins, 8 types of proteins S8, L15, L17, L21, L25, S7, SODa and Peptidylpropyl isomerase that are stably detected regardless of the strain and 4 types of proteins gns, YibT, YaiA and YciF, as biomarkers, their theoretical mass values were registered in the software as shown in Patent Literature 2.

(126) 5: 22962.8 that was within the mass difference of 800 ppm of SODa was registered as the closest 1: 22948.82, and 6: 22996.82 and 7: 23004.88 as 2: 23010.84. In addition, gns, YibT, YaiA and YciF in which contamination peaks exist are registered as 6483.51, 8023.08, 7110.89 and 18643.13/18653.16, respectively.

(127) Next, measured data in MALDI-TOF MS was analyzed with this software, and whether each biomarker was correctly attributed as a registered mass peak was examined. As a result, as shown in FIGS. 8A to 8G, all biomarker mass peaks of all the strains were attributed as registered mass numbers. Each attribution mass pattern was classified into groups 1 to 31, and compared with the serovar of each strain. Then, it was found that Typhimurium belongs to 1, 2 and 3, O4 group with unknown serovar to 4 and 5, Saintpaul to 6, O18 group with unknown serovar to 7, Orion to 8, Braenderup to 9, Montevideo and Schwarzengrund to 10, Schwarzengrund to 11, Abony and Pakistan to 12, Enteritidis to 13 and 14, Rissen to 15, *Gallinarum pullorum* to 16, Altona to 17, Amsterdam to 18, *infantis* to 19 and 20, Istanbul to 21, O4 group with unknown serovar to 22, Manhattan to 23, Mbandaka to 24, Senftenberg and O1, 3 and 19 groups with unknown serovar to 25, Thompson to 26, O4 group with unknown serovar to 27, O7 group with unknown serovar to 28, Brandenburg, Minnesota and Saintpaul to 29, Brandenburg and Saintpaul to 30, and *choleraesuis* strain to 31.

(128) Based on the above, it was found that use of the mass of S8 (m/z 13996.36 or 14008.41), L15 (m/z 14967.38, 14981.41 or 14948.33), L17 (m/z 14395.61 or 14381.59), L21 (m/z 11579.36 or 11565.33), L25 (m/z 10542.19 or 10528.17), S7 (m/z 17460.15, 17474.18 or 17432.1), SODa (m/z 22948.82, 23010.84, 22976.83 or 22918.79), Peptidylpropyl isomerase (m/z 10198.07 or 10216.11), gns (m/z 6483.51), YibT (m/z 8023.08), YaiA (m/z 7110.89) and YciF (m/z 18643.13) as biomarkers for MALDI-TOF MS analysis is useful for serovar identification of *Salmonella enterica* subsp. *enterica*.

(129) Among the biomarkers found out this time, 10 types except S8 and Peptidylpropyl isomerase have been reported in Non Patent Literature 10. However, Non Patent Literature 10 requires confirmation of each peak one by one, thus takes time for spectral analysis of MALDI-TOF MS for identifying serovar. Also, as to the mass-to-charge ratio m/z 6036 reported to be an important peak for identification of Enteritidis in Non Patent Literature 10, a peak was not confirmed in 5 strains out of 32 strains in Non Patent Literature 10, and in this example, a peak could not be confirmed in 8 strains out of 35 strains. Therefore, it was not used as a biomarker for serovar identification of *Salmonella enterica* subsp. *enterica*.

(130) By adding S8 and Peptidylpropyl isomerase to the biomarkers and using 12 types of carefully selected proteins as biomarkers, it became possible to provide a database that automatically identifies *Salmonella enterica* subsp. *enterica* to 31 groups for the first time.

(131) (6) Comparison with Fingerprint Method (SARAMIS)

(132) In fact, the identification result by the existing fingerprint method (SARAMIS) was compared with the identification result using the biomarker theoretical mass value shown in Table

6 as indices. First, in actual measurement in MALDI-TOF MS, a chart as shown in FIG. 9 was obtained. This result was analyzed by SARAMIS according to the instruction manual of AXIMA microorganism identification system. Results thus obtained are shown in FIG. 10A and FIG. 10B. As can be seen from these figures, all *Salmonella* genus bacteria used in the sample were identified as *Salmonella enterica* subsp. *enterica* in 91% to 99.9%, and species identification and serovar identification were not performed.

(133) Therefore, whether measurement results of strains of different subspecies can be identified based on the theoretical mass database shown in FIG. 8A was attempted. FIGS. 11 to 22 are enlarged views of 12 types of biomarker peak portions of the charts of FIG. 8. As can be seen from FIGS. 11 to 22, peaks can be distinguished since each biomarker mass is shifted. When compared with the measured values of 12 types of biomarkers and attributed, they agreed with the results shown in FIGS. 8A to 8D.

(134) Next, cluster analysis was performed using the attribution results of 12 types of ribosomal proteins, and dendrogram was generated. The results are shown in FIG. 23. In this method, although serovars of *infantis*, Brandenburg, Minnesota and Saintpaul could not be identified, other serovars could be almost identified.

(135) Based on the above, the following can be seen.

(136) SODa, S7 and gns are involved in the identification of multiple serovars and are particularly important as biomarkers for serovar identification of *Salmonella enterica* subsp. *enterica*.

(137) Moreover, Enteritidis, Mbandaka and *choleraesuis* can be identified from other serovars by combination of SODa and S7 mutation.

(138) Furthermore, *infantis* is identified, and Enteritidis and Mbandaka are identified by gns.

(139) Typhimurium, which is the top of serovar responsible for nontyphoidal *Salmonella* infections, is separated by YaiA, and Thompson by YibT. Also, Pullorum (*gallinarum*) is identified by L17, Rissen by S8, Orion by Peptidylpropyl isomerase, and Altona by L15. L25 separates *infantis* and Amsterdam, and L21 is important to identify Montevideo and Shwarzengrund, Minnesota. YciF is important for identification of *infantis*.

(140) (7) Gene Sequence and Amino Acid Sequence of Biomarkers

(141) DNA sequences and amino acid sequences in each strain of a total of 12 types of ribosomal proteins, S8, L15 and L17 encoded in the S10-spc-alpha operon and SODa, L21, L25, S7, gns, YibT, Peptidylpropyl isomerase and YciF outside the operon, which exhibit theoretical mass values different depending on the strain of *Salmonella enterica* subsp. *enterica*, are summarized in FIGS. 24 to 47.

REFERENCE SIGNS LIST

(142) 10 . . . Mass Spectrometry Unit 11 . . . Ionization Section 12 . . . TOF 13 . . . Extraction Electrode 14 . . . Detector 20 . . . Microorganism Discrimination Unit 21 . . . CPU 22 . . . Memory 23 . . . Display Section 24 . . . Input Section 25 . . . I/F 30 . . . Storage Section 31 . . . OS 32 . . . Spectrum Generation Program 33 . . . Genus/Species Decision Program 34 . . . First Database 35 . . . Subclass Decision Program 36 . . . Second Database 37 . . . Spectrum Acquisition Part 38 . . . m/z Reading Part 39 . . . Subclass Determination Part 40 . . . Cluster Analysis Part 41 . . . Dendrogram Generation Part

Claims

1. A method of identifying a serovar of *Salmonella* bacteria comprising a) a step of subjecting a sample containing microorganisms to mass spectrometry to obtain a mass spectrum, b) a step of reading a mass-to-charge ratio m/z of a peak derived from a marker protein from the mass spectrum, and c) an identification step of identifying a serovar of *Salmonella* bacteria in the sample, based on the mass-to-charge ratio m/z, wherein the serovars of *Salmonella* bacteria are classified using cluster analysis using as an index the mass-to-charge ratio m/z derived from at least

12 types of ribosomal proteins S8, L15, L17, L21, L25, S7, SODa, peptidylprolyl isomerase, gns, YibT, YaiA and YciF as the marker proteins.

2. The method according to claim 1, further comprising a step of generating a dendrogram representing an identification result by the cluster analysis.
